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ORIGINAL PAPER

Impact of electric vehicle development on China's energy consumption and greenhouse gas emissions

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Abstract

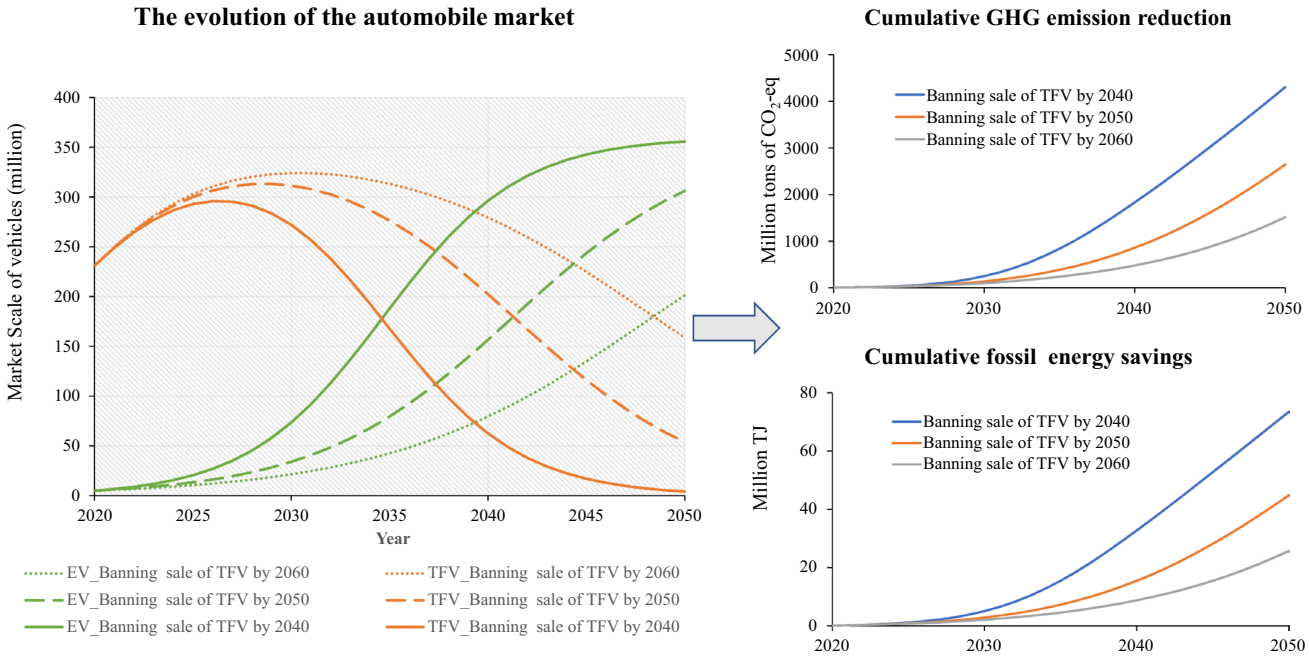
This study investigates the impact of electric vehicle development on China's greenhouse gas emissions and fossil energy consumption from a life cycle perspective. Based on vehicle technology and China's energy development plan, the potential for energy conservation and greenhouse gas emissions reduction of electric vehicles is explored. Utilizing a logistic model, the market evolution of electric vehicles under different timing for banning the sale of the traditional fuel vehicle is simulated and analyzed, and the corresponding effect on energy conservation and greenhouse gas emissions reduction is also examined. The results show that the high proportion of coal-fired thermal power limits the energy saving and emission reduction potential of electric vehicles in China. The life cycle greenhouse gas emissions and fossil energy consumption intensities of electric vehicles in various regions are approximately − 2.63%–30.98% (− 7.63–89.81 g CO_2-eq/km) and 17.71%–43.16% (0.67–1.64 MJ/km) lower than those of traditional fuel vehicles, respectively. As the proportion of renewable electricity generation continues to increase, by 2050, replacing traditional fuel vehicles with electric vehicles can reduce greenhouse gas emissions by at least 22.32% and fossil energy consumption by 29.94% in all regions of China. In addition, if China plans to ban the sale of traditional fuel vehicle between 2040 and 2060, the cumulative fossil energy savings and greenhouse gas emission reductions during the period of 2020–2050 can reach approximately 25.70–73.47 million TJ and 1511.95–4303.87 million tons of CO_2-eq, respectively.

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Graphic abstract



Keywords Battery electric vehicle · Ban sales of fossil fuel vehicle · Life cycle assessment · Energy consumption · Greenhouse gas emissions · Logistic equation

Introduction

Energy and environmental issues such as nonrenewable energy consumption and climate change are becoming major international issues that have a profound impact on global development in the twenty-first century (Kriegler et al. 2017; Brito et al. 2019). As the number of automobiles and their corresponding fuel consumption continue to increase, traditional fuel (e.g., gasoline and diesel) vehicles have become the major source of GHG emissions and fossil energy consumption around the world (Ramli et al. 2019; Hulshof and Mulder 2020; Matak and Krajačić 2020). With the sustained and rapid economic growth and the continuous improvement of people's living standards, China's car ownership has increased year by year, which has brought new pressures and challenges to China's energy and environment. At present, oil consumption in the automobile industry accounts for about half of China's total consumption, and 70% of the annual increase in oil consumption is used for new vehicles (State Statistical Bureau of China 2018; Sun and Wang 2018). However, China's net oil imports in 2016 were approximately 380 million tons, and its dependence on oil imports exceeded 60%, which poses a serious threat to the country's energy security and economic development (Zhang and Han 2017; State Statistical Bureau of China 2018). In addition, China's GHG emissions account for about 30% of the world's total GHG emissions, ranking first in the world, of which about 5% of the emissions are attributed to light-duty vehicles (Qiao et al. 2017; Sun and Wang 2018). With the rapid development of China's economy and society, the automotive market scale will continue to expand. Therefore, it is urgent to promote new energy vehicles (NEVs, also known as alternative fuel vehicles) to improve the sustainability of the energy and environment.

In the past decade, NEVs have received widespread promotion around the world (Plötz et al. 2016; Zhang et al. 2017; Watabe et al. 2020). Many countries and regions are actively promoting diversified alternative fuel vehicle plans, including the battery electric vehicle (BEV), hybrid electric vehicle (HEV), natural gas vehicle (NGV), and fuel-cell electric vehicle (FCEV) (Hao et al. 2017; Bicer and Dincer 2018; Shinde et al. 2021). The development of NEVs can not only effectively reduce the dependence of the transportation sector on fossil fuels, but also reduce air pollution and GHG emissions from the transportation sector (Krail and Schade 2016; Dong and Liu 2020). The development of the electric vehicle (EV) is considered a promising way to relieve China's energy and environmental pressures and has become the main direction of China's

new energy vehicle industry (Zhang et al. 2017; Zhao et al. 2019). In the past decade, China's EV market has experienced explosive growth. By 2020, the number of EVs in China has reached five million (China state information center 2021).

New energy vehicles have been extensively studied by scholars in various fields from different angles. Some have discussed the industrial planning and market prospects of NEVs (Jensen et al. 2017; Stangarone 2021), as well as the drivers, opportunities, and challenges of the development of NEVs (Berkeley et al. 2017; Gnann et al. 2018). Other scholars attended to the policy design, policy evaluation, and policy implications regarding the deployment of NEVs (Dong and Liu 2020; Samaie et al. 2020). The competitive advantage of NEVs over TFVs in terms of energy saving and environmental protection has been widely explored (Lang et al. 2013; Yang et al. 2016; Watabe et al. 2020). Moreover, a wealth of researches have reported on the factors influencing the deployment and adoption of NEVs such as economic, social, technological, and policy factors (Plötz et al. 2016; Egbue et al. 2017; Lévay et al. 2017), as well as the consumer's attitude, intention, and behavior (Singh et al. 2020; Haustein et al. 2021). The state of the art and trends of NEVs such as the components and architecture of vehicles, vehicle performance, and fuel and vehicle technology have been extensively studied (Singh et al. 2019, 2021; Ehsani et al. 2021).

However, there are few studies discussing the energy and environmental impact of a country's mid- and long-term deployment of new energy vehicles, especially the lack of analysis of the regional heterogeneity within a country. In addition, the actual energy saving and emission reduction effects of NEVs will depend on policy, market, and technology factors such as industrial strategy, energy planning, market scale of NEV, vehicle and fuel technology, etc. This complex and dynamic evolution process has often been neglected in previous studies. Moreover, some countries around the world have begun to draw up timetables for the withdrawal of traditional fuel vehicles from the market (i.e., plans to ban the sales of TFVs), ranging from 2025 to 2050 (Wappelhorst and Cui 2020), which will accelerate the transition from TFVs to NEVs and exert a complex and significant impact on the energy, environment, and related industries. Therefore, it is necessary to conduct a systematic analysis on the impact of banning the sale of TFVs on the entire automobile market and its consequent impact on energy and the environment so as to provide a reference for formulating reasonable and sustainable industrial policies and plans.

To supplement these gaps in the extant literature, this paper takes China as an example to simulate and analyze the market evolution of EVs and TFVs under different timings of banning the sale of TFVs based on a logistic model. Furthermore, the potential effect on fossil energy consumption and greenhouse gas (GHG) emission reduction was also examined. To the authors' knowledge, this is the first study to estimate the impacts of banning the sale of TFVs on the entire automobile market. In particular, combined with energy development planning and vehicle and fuel technology, it systematically and dynamically analyzes the impact of the market evolution of EVs on energy and the environment. The methodology used and the results presented can be used as a forward-looking reference by researchers, managers, authorities, and other relevant stakeholders to formulate more effective strategies to promote the deployment of NEVs and the sustainability of energy and the environment.

The remainder of this article is organized as follows. Section 2 lays out the materials and methods used in this study. Section 3 presents the results and discussion of the empirical analysis. Section 4 wraps up the paper with conclusions and recommendations, as well as limitations and future research directions.

Materials and methods

Life cycle assessment (LCA) is a standardized method used to measure the impact of products and services from "cradle to grave," covering the phases of resource extraction, manufacture, distribution, use, and disposal (ISO 2006a, b). This study aims to explore the impact of EV development on China's fossil energy consumption and GHG emissions from a life cycle perspective. The system boundary selected in this study includes fuel life cycle and vehicle life cycle (see Fig. 1). The life cycle assessment for fuels is known as well-to-wheels (WTW) analysis, which includes the measurement of energy consumption and emissions throughout the fuel life cycle from production, transportation, storage, and distribution to exhaust emissions (Ke et al. 2017; Rocco et al. 2018). As for electricity, its fuel life cycle mainly includes the stages of electricity production, transmission, battery charging, and use in vehicles. In addition, the vehicle life cycle (excluding the use of fuel) can be divided into stages of vehicle manufacture, vehicle maintenance, and end-of-life (i.e., vehicle scrapping and disposal) (ISO 2006a, b).

Currently, nearly 90% of the total number of automobiles in China are light-duty gasoline passenger cars; therefore, this study selected light-duty passenger vehicles as the research object for comparative analysis. Considering that BEVs account for more than 85% of China's total electric vehicles, and this proportion will continue to increase in the future; in this study, we chose BEV as the representative to study the impact of electric vehicle development on China's fossil energy consumption and GHG emissions. The materials and data used in this study were derived from the statistical reports of China's transportation and energy industries; peer-reviewed journal articles; and the Ecoinvent database,

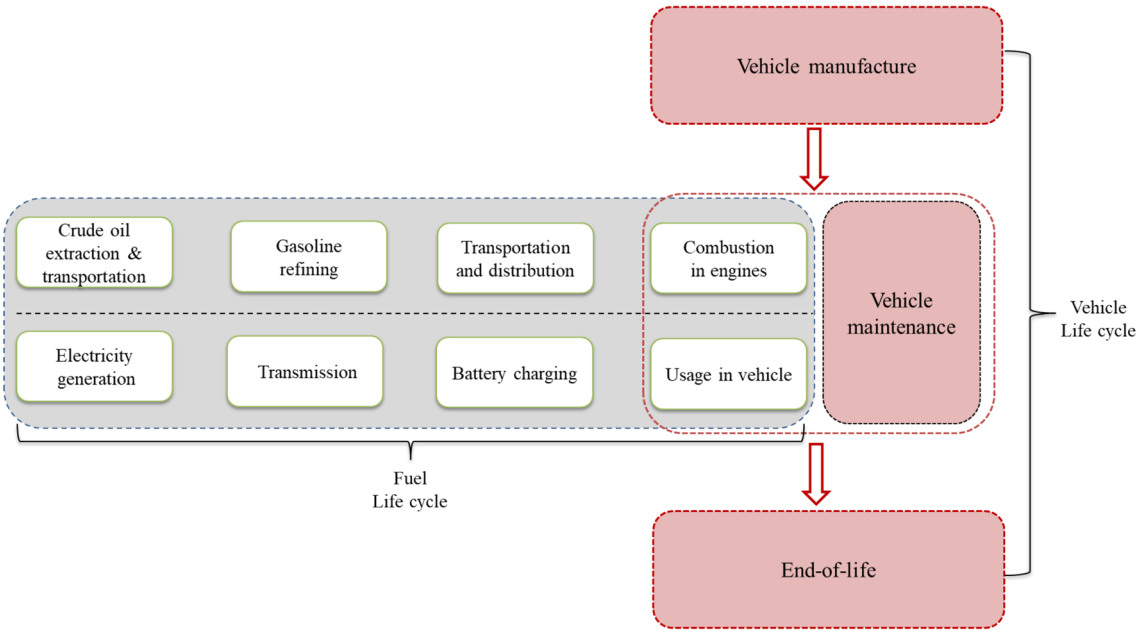


Fig. 1 System boundary of the life cycle assessment

Table 1 Specifications and mass of the TFV and EV (China Automotive Technology & Research Center 2020; China state information center 2021)

	TFV	EV
Components and mass		
Glider (kg)	926.25	1056.79
Powertrain (kg)	308.75	101.21
Lithium-ion battery (kg)	–	292.25
Gross vehicle weight (kg)	1235	1450.25
Fuel type	Gasoline	Electricity
Fuel economy (L/100 km; kWh/100 km)	7.8	17.3

Note: A glider is a vehicle without a powertrain

which is one of the most complete and used life cycle inventory databases in the world.

For evaluation, we selected the most common type of TFV and EV in China as the research objects, as shown in Table 1. TFVs can be subdivided into two modules, the glider and the powertrain; EVs include three modules, the glider, powertrain, and battery (Ecoinvent Centre, 2019). The lithium-ion battery mentioned in this study is the NMC battery (Lithium nickel manganese cobalt oxides, $LiNi_xMn_yCo_zO_2$), which is the most common type of EV battery used in China. The storage capacity and the mass of the battery are 40 kWh and 292.25 kg, respectively (Yang et al. 2020, 2021; Shi et al. 2021). Considering that the lifetime and life cycle driving kilometers of passenger cars in China are 10 years and 150,000 km, respectively, this study assumed that no battery replacement will be required in the vehicle life cycle (China Automotive Technology & Research Center 2020; China state information center 2021).

Regarding the market evolution of EV, this paper used a logistic function to simulate and predict the development trend of the vehicle market in China. The logistic function (also known as logistic curve), named by Pierre Francois Viller, is a common sigmoid function in the study of population growth (Nelder 1962; Morisita 1965). The generalized logistic curve can imitate the S-shaped curve of population growth in some cases (Nelder 1962; Morisita 1965). The logistic function has been widely used in the market diffusion of products and technologies and other fields of economics and sociology (Ramos 2013; Lotfi et al. 2014; Tarasov and Tarasova 2019). Therefore, in this study, we used the logistic equation in an attempt to simulate and predict the growth trend of China's automobile market. The parameter equation used in this study is shown in Eq. (1).

$$M(t) = \frac{A}{1 + Be^{-Ct}} \tag{1}$$

where A refers to the maximum capacity of China's automobile market, $M(t)$ refers to the car ownership at time t, and B and C are the two parameters of the logistic equation to be estimated.

Results and discussion

Potential energy saving and GHG emission reduction of EVs in China

To analyze the potential of EVs in reducing GHG emissions and fossil energy consumption, the functional units selected in this study are the life cycle fossil energy consumption and GHG emissions on the basis of per kilometer distance traveled. The life cycle fossil energy consumption intensity and GHG emission intensity of EVs and TFVs (i.e., gasoline-powered vehicles) in China can be calculated by Eqs. (2) and (3).

$$EC_j = ECVL_j + ECFL_j \tag{2}$$

$$GHG_j = GHGVL_j + GHGFL_j \tag{3}$$

where j refers to the vehicle type (i.e., EV and TFV), EC_j to the life cycle energy consumption intensity (MJ/km), GHG_j to life cycle GHG emission intensity (gCO_2-eq/km). *ECVL* and *ECFL* refer to the vehicle life cycle fossil energy consumption intensity and the fuel life cycle fossil energy consumption intensity, respectively. *GHGVL* and *GHGFL* refer to the vehicle life cycle GHG emission intensity and fuel life cycle GHG emission intensity, respectively. *ECVL* and *GHGVL* can be obtained by dividing the total energy consumption and total GHG emissions of the vehicle life cycle by total vehicle-kilometers traveled (VKT) during its lifespan. As for passenger cars in China, the average annual VKT is about 15,000 km and the average lifespan is about 10 years (China Automotive Technology & Research Center 2020; China state information center 2021).

Vehicle life cycle

For vehicle life cycle, data are mainly collected from the Ecoinvent database v. 3.6 (Ecoinvent Centre 2019). With regard to vehicle manufacture, the Ecoinvent "Passenger car, petrol {GLO}| market for | APOS, U" and "Passenger car, electric, without battery {GLO}| market for | APOS, U" datasets were used. Regarding vehicle maintenance, the "Maintenance, passenger car {GLO}| market for | APOS, U" dataset was used. As for the end-of-life stage, "Treatment of used glider, passenger car {GLO}| market for | APOS, U", "Treatment of used powertrain for passenger car, manual dismantling | APOS, U", and "Treatment of used Li-ion battery {GLO}| market for | APOS, U" datasets were used. Life cycle inventories for lithium-ion battery production in China are shown in Table 2.

The total GHG emissions and fossil energy consumption during the vehicle life cycle for TFV and EV are presented in Figs. 2 and 3.

As shown in Figs. 2 and 3, the total GHG emissions and fossil energy consumption of TFV's vehicle life cycle are about 10.75 tons of CO_2-eq and 110.84 GJ, respectively. For EV, the total GHG emissions and fossil energy consumption during the vehicle life cycle are approximately 16.01 tons of CO_2-eq and 190.58GJ, respectively. The results show that vehicle life cycle GHG emissions and fossil energy consumption of the EV are significantly higher than those of the TFV, which is mainly due to the high fossil energy consumption and GHG emissions of the battery. It can be found that the battery accounts for approximately 32.10% of GHG emissions and 40.24% of fossil energy consumption in an EV's life cycle.

Table 2 Life cycle inventories for lithium-ion battery production (Yin et al. 2019; Crenna et al. 2021)

Category	Material		Mass (kg)	Fossil energy consumption (MJ/kg)	GHG emissions (kg CO_2 -eq /kg)
Electrode active material	NMC		80.79	448.59	31.48
	Graphite		50.39	272.30	15.02
Electrode supportive material	Polyvinylidene fluoride		6.79	183.21	11.47
	N-Methyl-2-pyrrolidone		108.81	82.09	2.48
	Al foil		11.20	288.16	17.87
	Cu foil		26.02	45.09	3.01
Separator	Polyethylene membrane		3.61	149.22	8.84
Electrolyte material	Lithium hexafluorophosphate		5.22	360.13	26.43
	Ethylene carbonate		14.01	34.08	1.65
	Dimethyl carbonate		14.01	47.60	1.31
Packaging	Cu tab	8.79		27.88	3.42
	Al housing	4.82		254.35	15.75

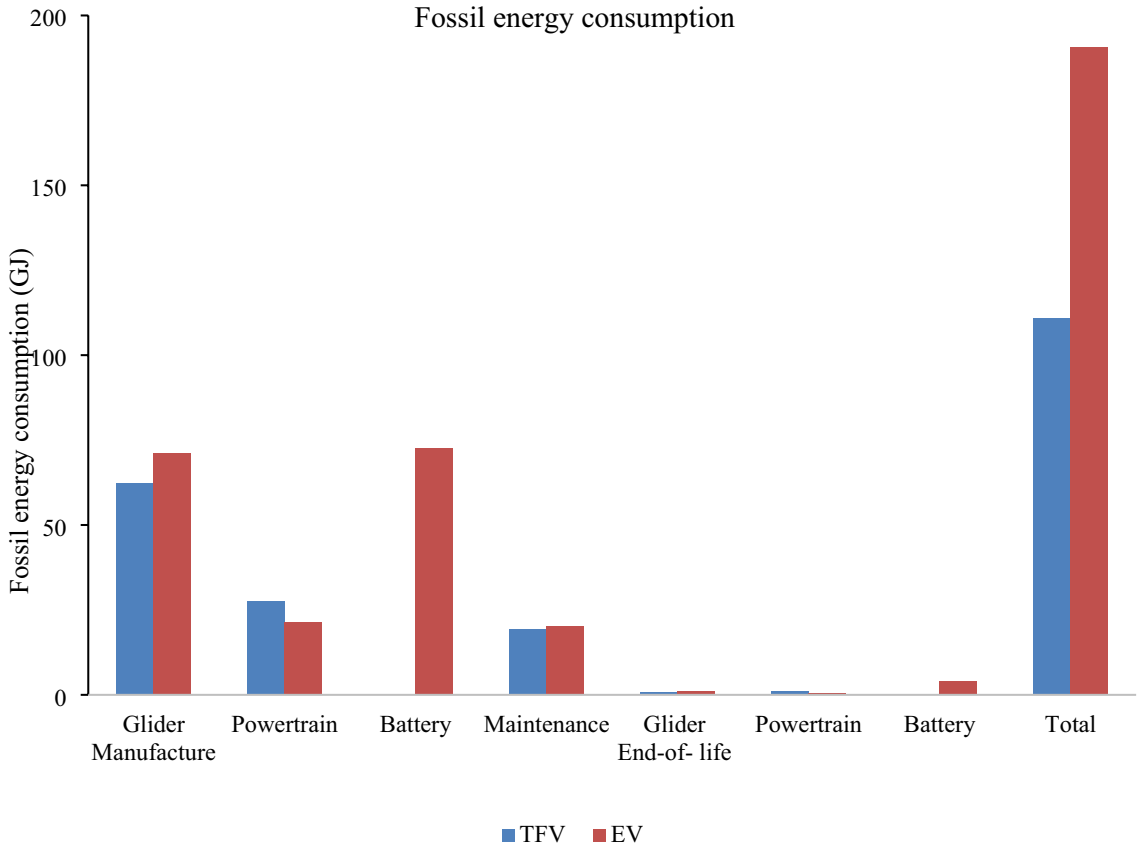


Fig. 2 Fossil energy consumption during the vehicle life cycle

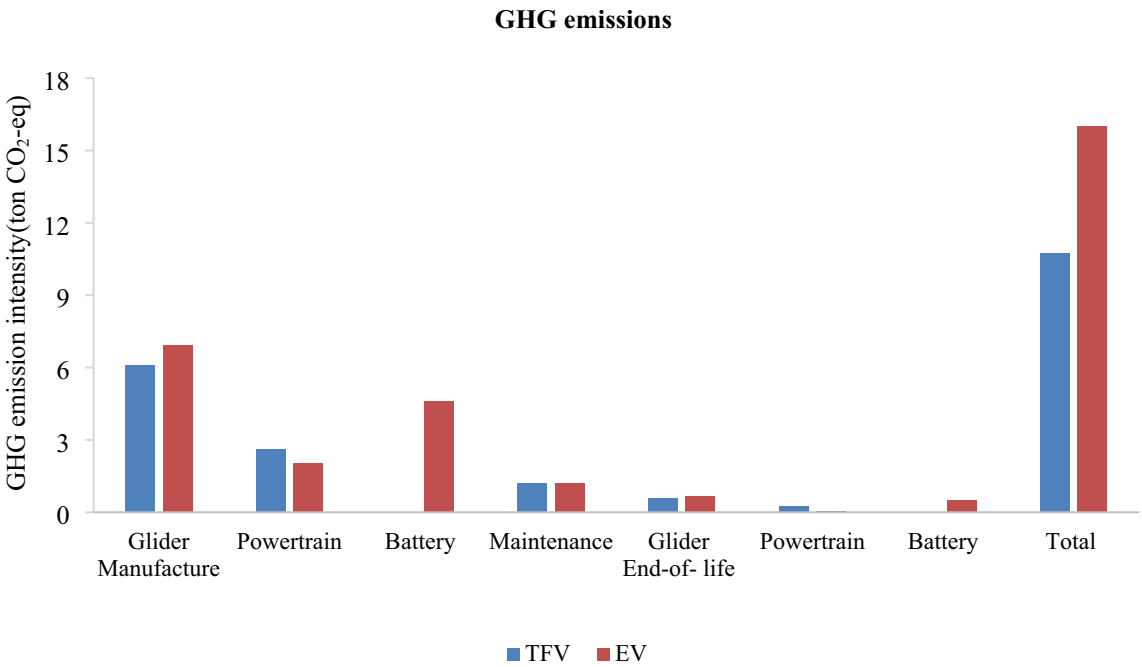


Fig. 3 GHG emissions during the vehicle life cycle

Fuel life cycle

The fuel life cycle energy consumption and GHG emissions of a vehicle are mainly determined by its fuel economy and the life cycle GHG emission intensity of the fuel. As for gasoline-powered vehicles, the fuel life cycle fossil energy consumption intensity and GHG emission intensity can be calculated by Eqs. (4) and (5), respectively.

$$ECFL_{GV} = ECG \times FE_{GV} \tag{4}$$

$$GHGFL_{GV} = GHGG \times FE_{GV} \tag{5}$$

where $GHGFL_{GV}$ refers to the fuel life cycle GHG emission intensity of gasoline-powered vehicles(g CO_2-eq/km); $ECFL_{GV}$ refers to the fuel life cycle energy consumption intensity of the gasoline-powered vehicles (i.e., MJ/km); FE_{GV} refers to the fuel economy of the gasoline-powered vehicles (i.e., L/100 km); and *ECG* and *GHGG* refer to the life cycle energy consumption intensity and GHG emission intensity of the gasoline in China, respectively, which are about 1.268 MJ/MJ and 2.797 kg CO_2-eq/L (National Bureau of Statistics of China 2020).

Although the promotion and use of EV can directly reduce fossil fuel consumption, the production of electricity in China requires the consumption of fossil energy. Therefore, when analyzing the energy saving and emission reduction potential of EV, it is necessary to fully consider China's power production. The fuel life cycle fossil energy consumption intensity and GHG emission intensity for an EV can be calculated by Eqs. (6) and (7), respectively.

$$ECFL_{EV} = FE_{EV} \times \beta^{-1} \times \partial^{-1} \times ECE \tag{6}$$

$$GHG_{EV} = FE_{EV} \times \beta^{-1} \times \partial^{-1} \times GHGE \tag{7}$$

where $ECFL_{EV}$ refers to the fuel life cycle energy consumption intensity of EV (MJ/km); $GHGFL_{EV}$ refers to the fuel life cycle GHG emission intensity of EV (g $CO_{2\text{-eq}}$/km); ∂ refers to the power transmission and distribution efficiency of the power network; and β is the charging efficiency of the EV's battery. According to the China Automotive Technology & Research Center (2020) and China National Energy Administration (2020), the value of ∂ and β are 93% and 90%, respectively. FE_{EV} refers to the fuel economy of EV (i.e., kWh/100 km), *ECE* refers to the life cycle fossil energy consumed to provide 1 kWh of electricity in China (MJ/kWh), and *GHGE* refers to the life cycle GHG emissions from 1 kWh of electricity (g CO_2-eq/kWh).

Currently, the *ECE* and *GHGE* of thermal power in China are 9.96 MJ/kWh and 1021.35 g CO2-eq/kWh, respectively (Intergovernmental Panel on Climate Change (IPCC) 2006; China National Energy Administration 2020). In addition, for nuclear power, the *ECE* and *GHGE* are 0.23 MJ/kWh and 23.42 g CO_2-eq/kWh, respectively (China National Energy Administration 2020; National Bureau of Statistics of China 2020). For wind power generation, the *ECE* and *GHGE* are 0.17 MJ/kWh and 7.20 g CO2-eq/kWh, respectively (China National Energy Administration 2020; National Bureau of Statistics of China 2020). For solar power generation, the *ECE* and *GHGE* are 0.44 MJ/kWh and 62.10 g CO2-eq/kWh, respectively (China National Energy Administration 2020; National Bureau of Statistics of China 2020). For hydropower, the fossil energy consumption can be negligible, while GHG emissions are approximately 18.15 g CO_2-eq/kWh (China National Energy Administration 2020; National Bureau of Statistics of China 2020).

As for the current fuel economy of passenger vehicles in China, according to the China Automotive Technology & Research Center (2020) and Ministry of Industry and Information Technology of China (2021), the average fuel economy of an EV is 17.3 kWh/100 km, while the gasoline consumption rate of gasoline-powered vehicles is about 7.8 L/100 km. The life cycle fossil energy consumption and GHG emission intensities of different vehicle fuel pathways (electricity for EV) in China are shown in Figs. 4 and 5.

As shown in Figs. 4 and 5, the life cycle GHG emission intensity and fossil energy consumption intensity of TFV are approximately 289.87 g CO_2-eq/km and 3.81 MJ/km, respectively. Among them, the fuel life cycle accounts for about 80.59% of fossil energy consumption and 75.27% of greenhouse gas emissions, while the vehicle life cycle accounts for about 19.41% of fossil energy consumption and 24.73% of GHG emissions. Compared with the TFV, the EV powered by electricity produced from renewable energy sources has significant potential for energy saving and GHG emission reduction; it can reduce GHG emission intensity and fossil energy consumption intensity by approximately 61% (177.10 g CO_2-eq/km) and 65% (2.49 MJ/km), respectively.

However, an EV driven by coal-fired thermal power has no advantages in terms of energy saving and emission reduction. Its life cycle fossil energy consumption intensity is only 9.66% (0.37 MJ/km) lower than that of the TFV. In particular, the life cycle GHG emission intensity of an EV driven by coal-fired thermal power is approximately 13.49% higher than that of TFV (39.09 g CO2-eq/km).

Therefore, only when coal-fired thermal power is partially replaced by renewable energy power can EV have a lower GHG emission intensity than TFV. Specifically, in China, to realize the GHG emission reduction potential of EV, the proportion of renewable energy power generation in the power supply mix should exceed about 19%.

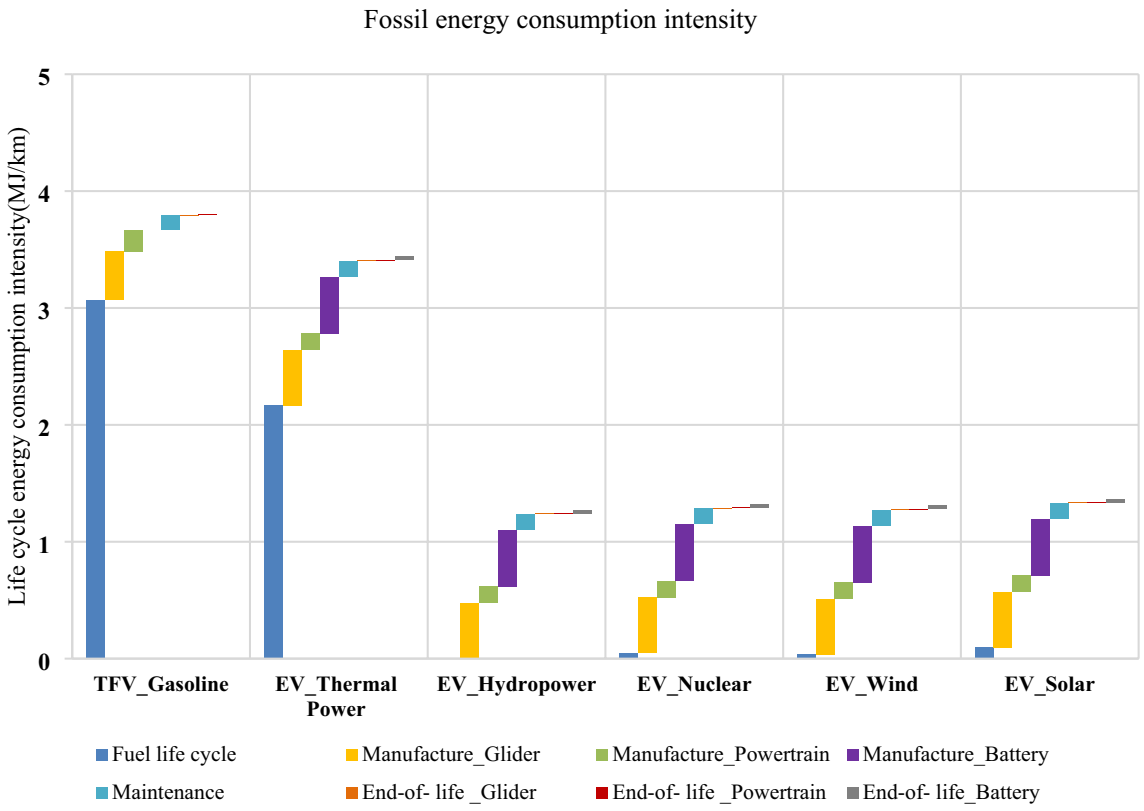


Fig. 4 Life cycle fossil energy consumption intensities of different vehicle fuel pathways

Energy saving and emission reduction potential under China's current electricity mix

Due to the different power structures in different regions of China, the GHG emission reduction potential of EV is also heterogeneous. Under the current electricity generation mix for regional grids in China, *ECE* and *GHGE* can be calculated by Eqs. (8) and (9), respectively.

$$ECE = \sum ECE_i \times S_i \tag{8}$$

$$GHGE = \sum GHGE_i \times S_i \tag{9}$$

where i refers to the category of the energy source for electricity production in China (i.e., Thermal power, Hydropower, Nuclear, Wind, Solar) and S_i denotes the proportion of category i electricity production to the total power generation. China's current electricity generation mix is shown in Fig. 6.

It can be seen from Fig. 6 that coal-fired thermal power is the primary source of electricity supply in China, and the proportion of electricity generated from renewable energy is still low (China National Energy Administration 2020). In South, East, Central, Northwest, Northeast, and North China, coal-fired thermal power accounts for 40.89%, 81.33%, 66.38%, 69.46%, 74.56%, and 85.52%, while renewable energy power generation accounts for 55.11%, 18.67%, 33.62%. 30.54%, 25.44%, and 14.48%, respectively. According to Eqs. (2)–(9), the life cycle fossil energy consumption and GHG emission intensities of EVs and TFVs under the current electricity generation structure of China's regional power grids can be obtained, as shown in Fig. 7.

As can be seen from Fig. 7, there are significant differences in the GHG emission reduction and energy saving effects of EVs in various regions of China. In South China, due to the high proportion of renewable energy power generation (55.11%), EVs can exert a strong energy saving and emission reduction effect. Its life cycle, GHG emissions and fossil energy consumption intensities are approximately 30.98% (89.81 g CO_2-eq/km) and 43.16% (1.64 MJ/km) lower than those of TFV, respectively. In the northeast, northwest, and central regions, the life cycle GHG emission intensity of EVs is about 5.63% (16.31 g CO_2-eq/km), 9.39% (27.22 g CO2-eq/km), and 11.78% (34.14 g CO2-eq/km) lower than that of TFVs, respectively. In these regions, the fossil energy consumption intensity of EVs is also about 23.86%–28.71% (0.91–1.09 MJ/km) lower than that of TFVs.

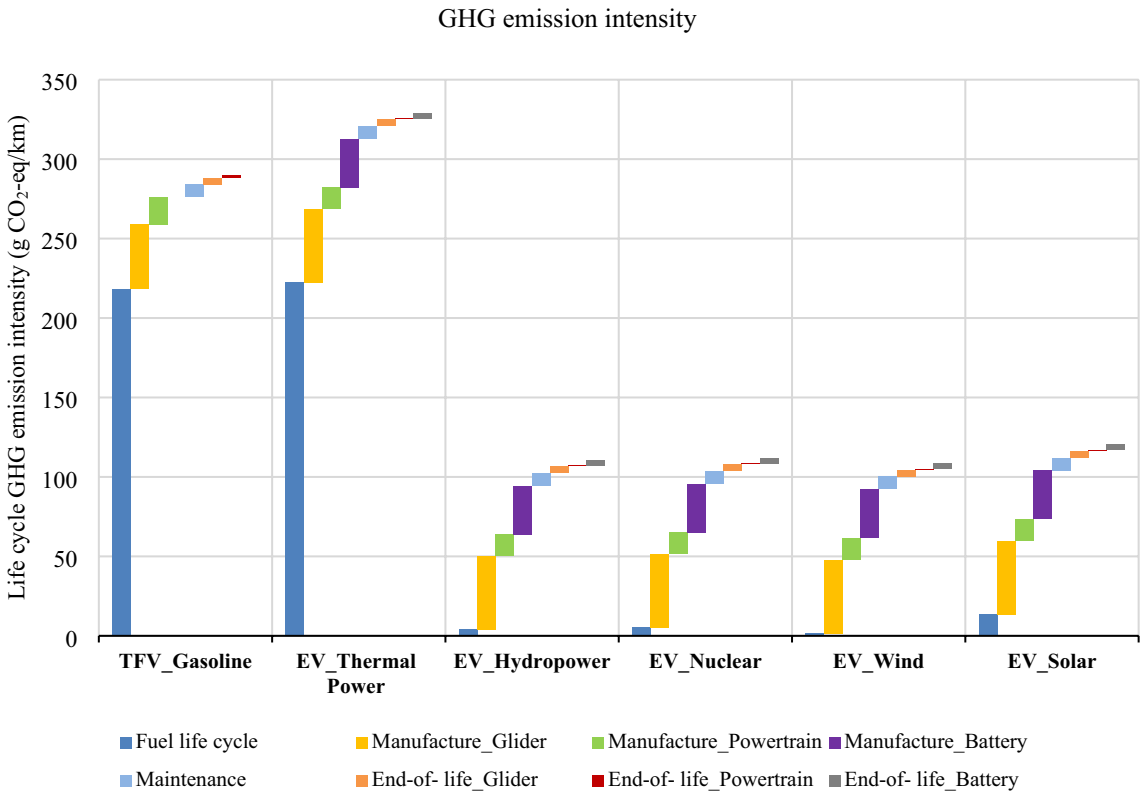


Fig. 5 Life cycle GHG emission intensities of different vehicle fuel pathways

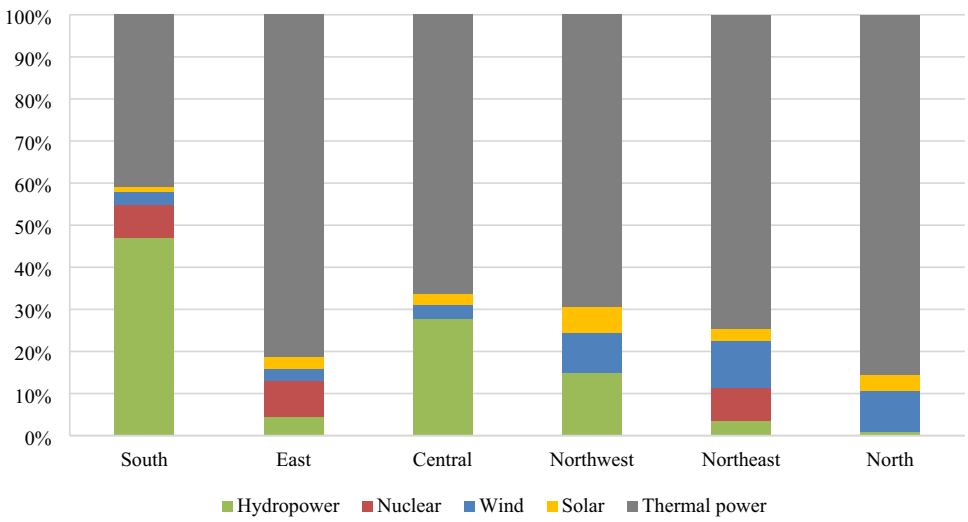


Fig. 6 Electricity generation mix for China's regional power grids in 2020 (China National Energy Administration 2020)

However, in North China and East China, due to the large proportion of thermal power,

EV has no advantage over TFV in terms of reducing GHG emissions. Especially in North China, thermal power accounts for 85.52% of the entire power supply network, making the EV's life cycle GHG emission intensity slightly greater than that of the TFV.

Energy saving and emission reduction potential under China's energy planning scenario

To cooperate with the development of EVs, China has formulated relevant energy development plans to better ease the pressure on energy and environment. According to China's power development planning, the proportion of

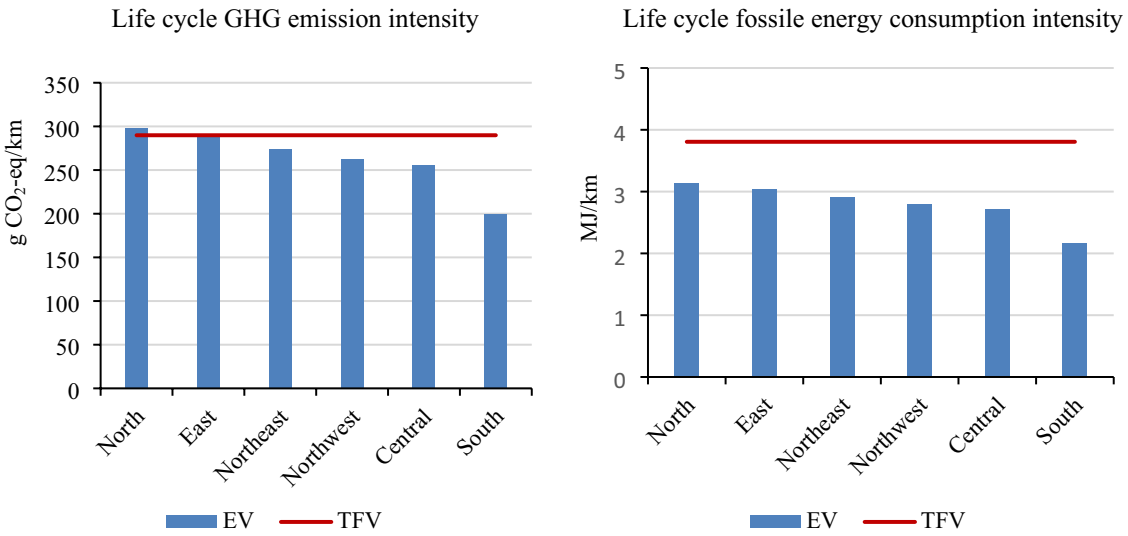


Fig. 7 Fossil energy consumption and GHG emission intensities under current electricity generation structure

coal-based thermal power generation is expected to drop to about 60% in 2030 and 50% in 2050 (National Development and Reform Commission 2016; Elshkaki 2019), as shown in Fig. 8.

At the same time, the fossil energy consumption of the coal-based thermal power generation is expected to drop to 8.94 MJ/kWh in 2030 and 8.06 MJ/kWh in 2050 (China Petroleum Economics and Technology Research Institute 2019; National Bureau of Statistics of China 2020).

In addition, the fuel economy of EVs on the road in China is expected to be 10 kWh/100 km in 2030 and 6 kWh/100 km in 2050 (China Automotive Technology & Research Center, 2020; Government of China, 2020), and the fuel economy of the light-duty gasoline-powered vehicles on the road in 2030 and 2050 is expected to be 6 L/100 km and 4 L/100 km (China Automotive Technology & Research Center 2020; China state information center 2021).

Based on the above energy planning and assumption, we further compared and analyzed the fossil energy consumption intensity and the GHG emission intensity of EVs and TFVs in the future, as shown in Figs. 9 and 10.

As can be seen from Figs. 9 and 10, with the increase in the proportion of renewable energy power generation and the decrease in energy consumption for power generation in

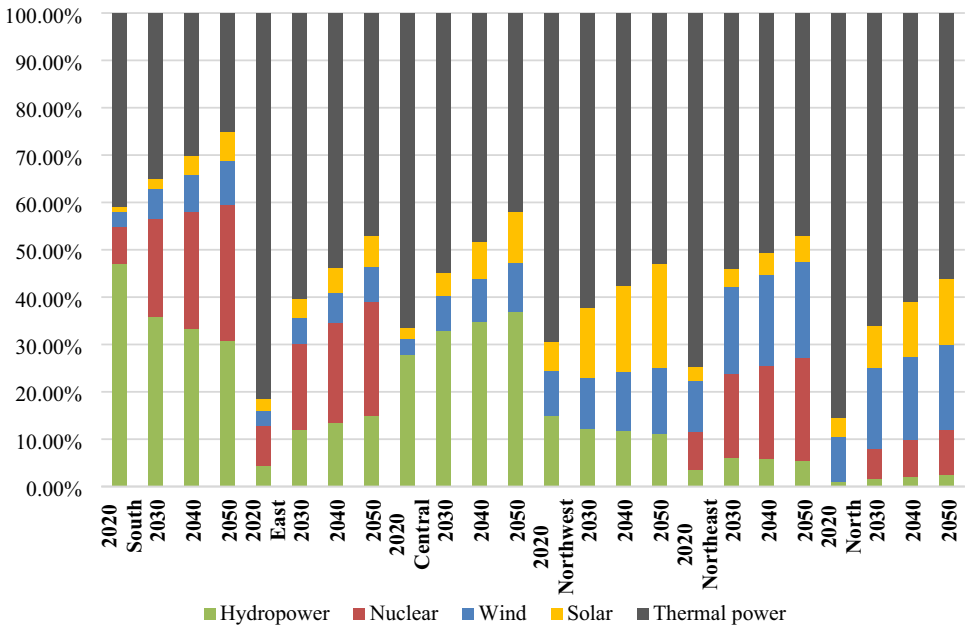


Fig. 8 China's regional electricity generation plan (China National Energy Administration 2020; China Petroleum Economics and Technology Research Institute 2019)

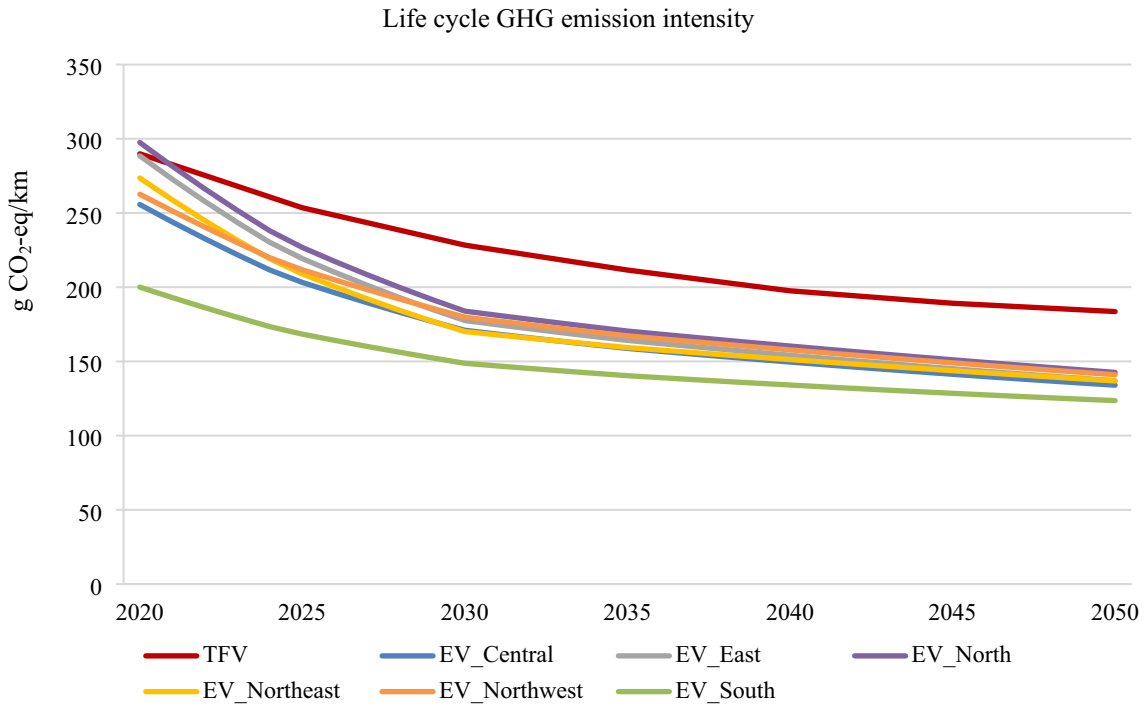


Fig. 9 Comparison of GHG emission intensity under China's power development scenario

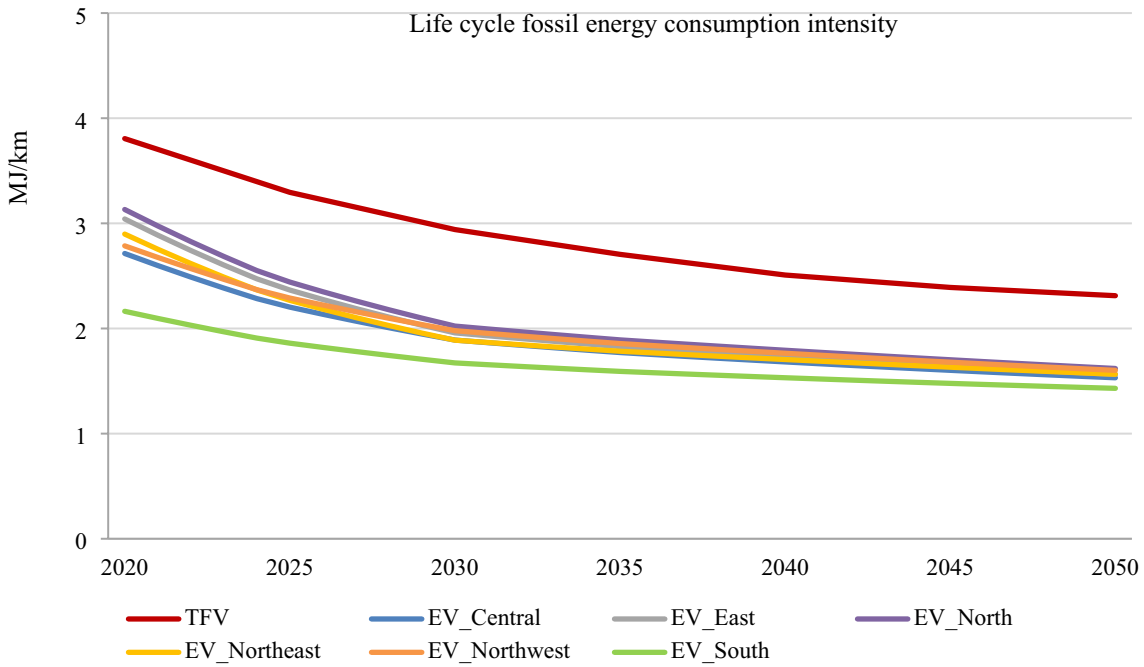


Fig. 10 Comparison of fossil energy consumption intensity under China's power development scenario

coal-fired thermal power plants, the advantages of EVs in energy conservation and emission reduction will continually become more significant. In South China, the life cycle GHG emission intensity of EV will drop to 148.72 g CO2-eq/km in 2030 and 123.60 g CO2-eq/km in 2050, while the fossil energy consumption intensity of EV could also be reduced to 1.67 MJ/km in 2030 and 1.43 MJ/km in 2050. For other regions, the GHG emission intensity of EV will drop to

approximately 176 g CO2-eq/km in 2030 and 138 g CO2-eq/km in 2050, while the fossil energy consumption intensity of EV could be reduced to 1.95 MJ/km in 2030 and 1.58 MJ/km in 2050. By then, in these regions, replacing TFVs with EVs can reduce life cycle GHG emissions and fossil energy consumption by more than 25% and 32%, respectively.

Market evolution of automobiles in China

Considering that the growth of car ownership has certain logistic growth curve characteristics, we attempt to simulate and predict the evolution trend of China's automobile market. The value of the parameters in Eq. (1) were obtained by using historical data of car ownership in China from 1985 to 2018. This paper uses the nonlinear least squares method (NLS) to obtain the values of 3 parameters in the objective function. The initial values of A, B, and C are set to 45,000 (ten thousand vehicles), 200, and 0.2, respectively. The iterative algorithm is the classic McQuartt method. Software SPSS 26.0 was used to perform the entire solution process. The results are shown in Table 3. It can be seen that the goodness of fit of the logistic equation is strong, and the trend of China's car ownership can be obtained by using this estimated equation (see Fig. 11). In addition, some simulation values and the actual values were selected for comparison. The results are shown in Table 4.

It can be seen from Table 4 that the errors between the simulation values and the actual values from 2016 to 2019 are all within 5%. Therefore, the logistic equation can effectively simulate the growth trend of China's automobile market. As shown in Fig. 11, China's car ownership will gradually reach a saturation state after 2030, and the saturation value is about 357 million vehicles. Currently, some countries around the world have begun to draw up timetables for the withdrawal of traditional fuel vehicles from the market (i.e., plans to ban the sale of TFVs), ranging from 2025 to 2050 (Wappelhorst and Cui 2020). The Chinese government does not have a clear plan yet; however, considering the actual situation in China, it is expected that the ban will be set at approximately 2050. Considering that the service life of passenger cars in China is about 10 years (China Automotive Technology & Research Center 2020; China state information center 2021), it can be assumed that EVs will account for more than 90% of the overall China automobile market 10 years after banning the sale of TFVs. In addition, the market scale of EVs in China was about 5 million in 2020 (China Automotive Technology & Research Center 2020; China state information center 2021). Based on these data and assumptions, the market evolution of EVs and TFVs under different timings of banning the sale of TFVs can

Table 3 The logistic equation for China's automobile market

Logistic equation	Statistical result
Parameters	
A	35,691.60*** (0.001)
B	3561.16*** (0.000)
C	0.25*** (0.000)
Fit statistics	
R^2	0.986
Adjusted R^2	0.951

Note: P-values are in parentheses. Significance levels: * $p<0.05$; ** $p<0.01$; *** $p<0.001$

Table 4 The comparison and error between the actual value and the simulation value

Year	Actual value	Simulation value	Error
2016	16,599.00	16,831.11	1.39%
2017	18,469.54	19,076.62	3.29%
2018	20,555.40	21,283.65	3.54%
2019	22,504.50	23,186.61	2.94%

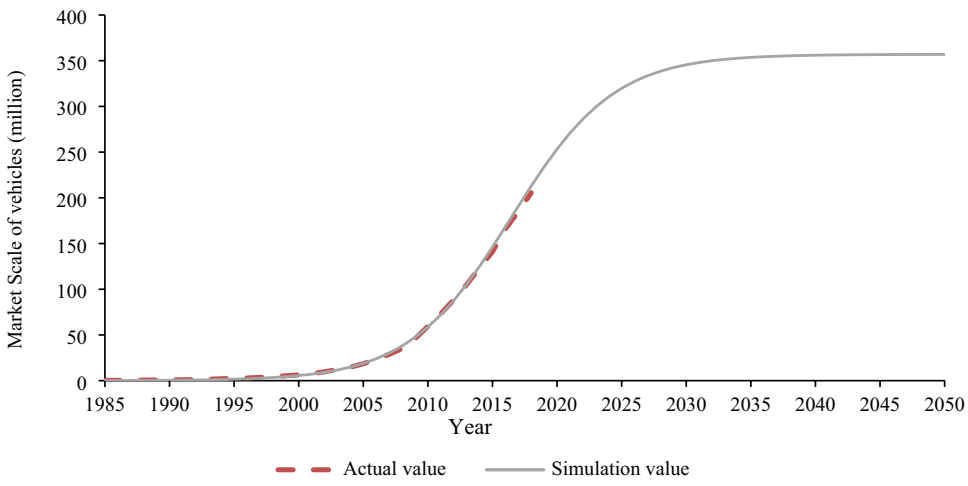


Fig. 11 The trend of China's automobile market

be obtained by using logistic equation. The detailed results are shown in Fig. 12.

As shown in Fig. 12, if China bans the sale of TFVs from 2060, the market scale of EVs will exceed 100 million by approximately 2042 and surpass the market scale of TFVs by 2049. If China bans the sale of TFVs from 2050, the number of EVs will reach 100 million by approximately 2037 and surpass the market scale of TFVs by approximately 2041. Particularly, if China plans to ban the sale of TFVs from 2040, the market scale of EVs will exceed that of TFVs by approximately 2035.

Effect of energy saving and GHG emission reduction under different development scenarios of EV

Based on the fossil energy consumption and GHG emission intensities of EVs and TFVs, and the market evolution of automobile in China, we can estimate the effect of energy saving and GHG emission reduction under different development scenarios for EVs.

The annual total GHG emissions (AT_{GHG}) and annual total energy consumption (AT_{EC}) of China's automobile market can be calculated by Eqs. (10) and (11).

$$AT_{GHG} = \sum NV_{j,k} \times VKT \times GHG_{j,k} \tag{10}$$

$$AT_{EC} = \sum NV_{j,k} \times VKT \times EC_{j,k} \tag{11}$$

where j is the vehicle type (i.e., EV and TFV), k is the region (i.e., South, Northeast, Northwest, Central, East, and North China), NV is the number of vehicles, and VKT is the average annual vehicle kilometer traveled. EC_j and GHG_j refer to the life cycle energy consumption intensity (MJ/km) and life cycle GHG emission intensity (gCO_2-eq/km), respectively, which can be calculated by Eqs. (2)–(9). Considering that cars can stay on the road for years (i.e., service life of the vehicle) and the fuel economy (FE) is expected to change drastically over that period, this study used the weighted moving average method to calculate the FE_t of each type of vehicle in each year, as shown in Eq. (12).

$$FE_t = \frac{FE_{t-1} \times \left(NV_{t-1} - \frac{NV_{t-1}}{LS}\right) + FE_t \times \left(NV_t - NV_{t-1} + \frac{NV_{t-1}}{LS}\right)}{NV_t} \tag{12}$$

where t is the time (i.e., year) and LS is the lifespan of vehicle. For each year, the newly added vehicles ($NV_t - NV_{t-1}$) and the vehicles replaced due to the end of their service life (NV_{t-1}/LS) will adopt the latest FE. To investigate the energy and environment impacts of the medium- and long-term deployment of EV, we first calculated and compared the annual GHG emissions and energy consumption of the entire China automobile market with and without EV, and then used 2020 as the starting point to calculate the cumulative fossil energy savings and GHG emission reductions

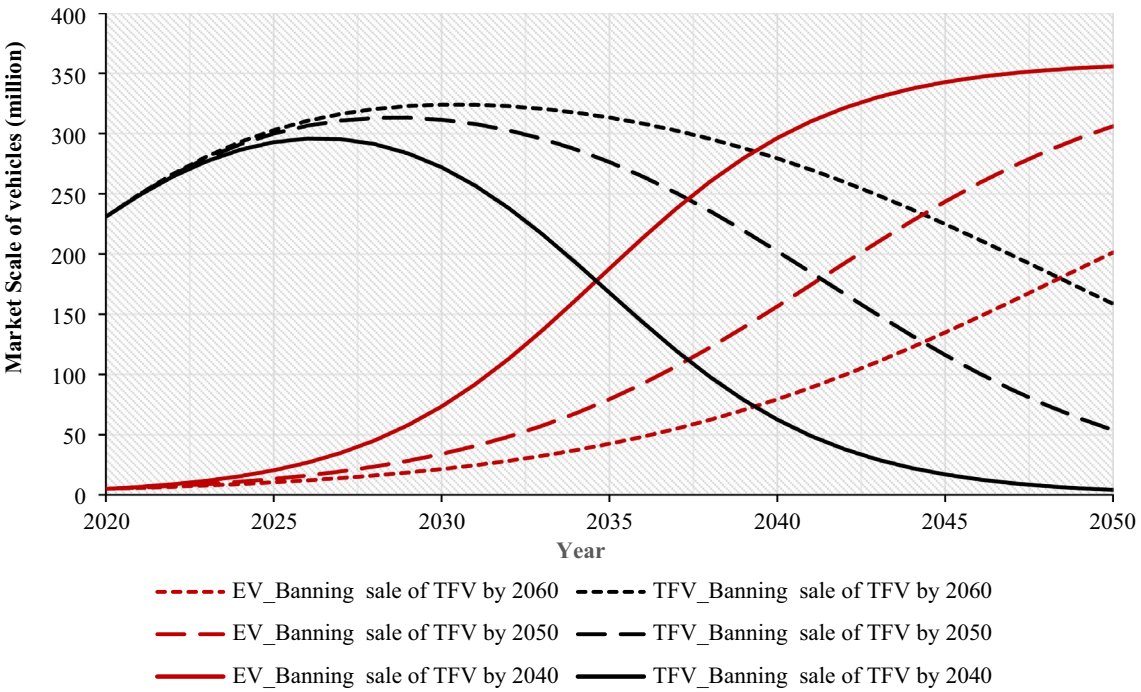


Fig. 12 Development trends of the EV and TFV markets in China

brought about by the promotion of EV during the period of 2020–2050. The detailed results are shown in Fig. 13.

It can be seen from Fig. 13 that the overall effect of energy saving and emission reduction before 2030 is small, which is mainly due to the high proportion of thermal power and the small market scale of EVs during this period. After then, with the continuous increase in the proportion of renewable energy electricity and the scale of EVs, the GHG emissions and fossil energy consumption of the automobiles market in China can be significantly reduced.

If China plans to ban the sales of TFVs from 2060, a cumulative fossil energy savings of 2.07 million TJ and a cumulative GHG emission reduction of 96.42 million tons of CO_2-eq can be achieved by 2030. By 2050, the cumulative energy saving and GHG emission reduction can reach 25.70 million TJ and 1511.95 million tons of CO_2-eq, respectively.

If sales of TFVs are banned from 2050, the cumulative energy savings can reach 2.83 million TJ by 2030 and 44.82 million TJ by 2050. In addition, the cumulative GHG emission reduction can also reach 134.92 million tons of CO_2-eq by 2030 and 2644.58 million tons of CO_2-eq by 2050.

In particular, if China plans to ban the sale of TFVs by 2040, the cumulative energy savings and GHG emission reduction can reach 5.04 million TJ and 247.23 million tons of CO_2-eq by 2030. By 2050, the cumulative energy savings and GHG emission reduction can reach 73.47 million TJ and 4303.87 million tons of CO_2-eq, respectively.

Conclusions

This study explores the impact of promoting EVs on fossil energy consumption and GHG emissions in China from a life cycle perspective. Combined with the development of China's energy sector and vehicle technology, the comparative advantages of EV over traditional gasoline-powered vehicle in terms of energy saving and GHG emission reduction were analyzed. In addition, the logistic equation was used to simulate and predict the growth trend of EVs with different timings of banning the sale of TFVs. Finally, on the basis of the market scale of EVs and TFVs in China, the impact of the medium- and long-term development of EV on energy and the environment was further estimated. The main conclusions are presented as follows.

The vehicle life cycle GHG emissions and fossil energy consumption of EVs are significantly higher than those of TFVs, which is mainly due to the high fossil energy consumption and GHG emissions of the battery. The lithium-ion battery accounts for approximately 32.10% of GHG emissions and 40.24% of fossil energy consumption in an EV's life cycle. From the perspective of a full life cycle (i.e., vehicle life cycle plus fuel life cycle), the EV powered by the electricity produced from renewable energy sources has significant potential in energy saving and GHG emission reduction. It can reduce GHG emission intensity and fossil energy consumption intensity by approximately 61% and 65%, respectively. However, the life cycle GHG emission intensity of EV driven by coal-fired thermal power is approximately 13.49% higher than that of TFV. Under China's current electricity generation technologies, to realize the GHG emission reduction potential of EVs, the proportion of

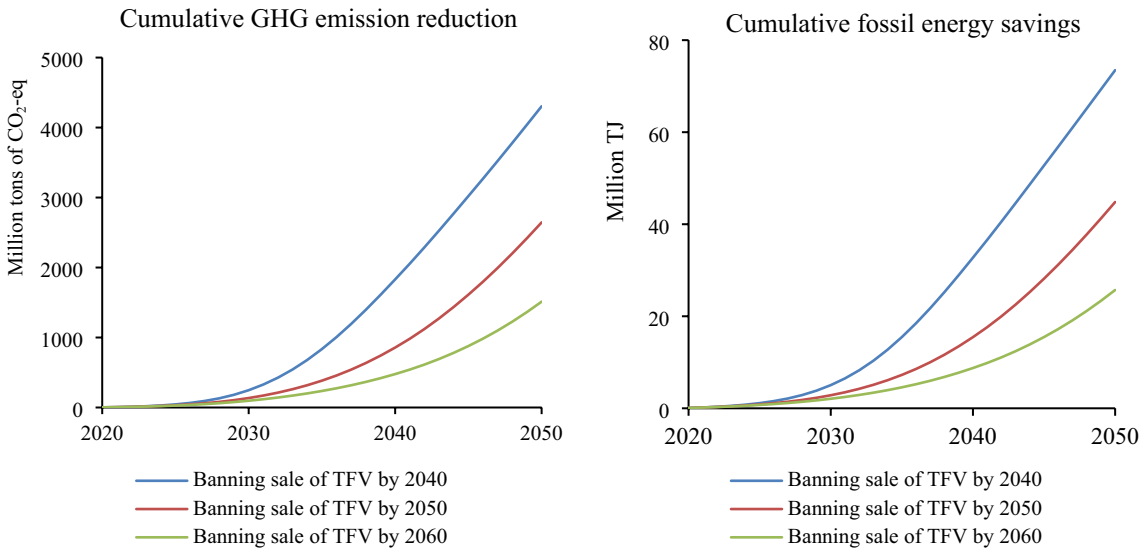


Fig. 13 Cumulative GHG emission reduction and fossil energy savings

renewable energy power generation in the power supply mix should exceed 19%.

There are significant differences in the GHG emission reduction and energy saving effects of EVs in various regions of China. In South China, due to the high proportion of renewable energy power generation (55.11%), the life cycle GHG emissions and fossil energy consumption intensities of EVs are approximately 30.98% and 43.16% lower than those of TFVs, respectively. For other regions with a high proportion of coal-fired thermal power generation, such as the northeast, northwest, and central regions of China, the full life cycle GHG emission and fossil energy consumption intensities of EVs are only about 0.47%–11.78% and 20.08%–28.71% lower than those of TFVs, respectively. Especially in North China, thermal power accounts for 85.52% of total electricity generation, making the EV's life cycle GHG emission intensity slightly greater than that of the TFV. As the proportion of China's renewable electricity generation continues to increase, the advantages of EV in energy conservation and emission reduction will become more and more significant. By 2050, replacing TFVs with EVs can reduce life cycle GHG emissions and fossil energy consumption by more than 25% and 32%, respectively.

If China bans the sale of TFVs from 2060, the market scale of EV will exceed 100 million by approximately 2042 and surpass the market scale of TFV by 2049. During 2020–2050, the cumulative fossil energy savings and GHG emission reductions can reach 25.70 million TJ and 1511.95 million tons of CO_2-eq, respectively. If China bans the sale of TFVs from 2050, the number of EVs will reach 100 million by approximately 2037 and surpass the market scale of TFVs by approximately 2041. During the period of 2020–2050, a total of approximately 44.82 million TJ fossil energy savings and 2644.58 million tons of CO_2-eq GHG emission reductions can be obtained. In particular, if China plans to ban the sale of TFVs from 2040, the market scale of EVs will exceed that of TFVs by approximately 2035. During 2020–2050, the cumulative fossil energy savings and GHG emission reductions can reach approximately 73.46 million TJ and 4303.87 million tons of CO_2-eq, respectively.

Nowadays, China faces increasing pressure regarding energy and environment in the process of economic development. The promotion of electric vehicles is one of the potential ways to effectively alleviate these problems. Banning the sale of traditional fuel vehicles is a long-term plan that requires comprehensive consideration of many factors such as economy, market, technology, energy, and environment. Authorities should make reasonable plans and adequate preparations in advance to promote a healthy and smooth transition from traditional fuel vehicles to electric vehicles. Firstly, the large-scale promotion of electric vehicles is bound to have a significant impact on the traditional automobile industry and the fossil energy industry; therefore, the transformation and development of the existing automobile industry and the fossil energy industry should be fully considered. Secondly, the charging infrastructure and power supply capacity are important factors that affect the large-scale promotion of electric vehicles. Therefore, the construction of charging infrastructure and the transformation and upgrading of power grid should be arranged in advance to provide a favorable supporting environment for the application of electric vehicles. In addition to policy factors, the market competitiveness of electric vehicles relative to traditional fuel vehicles is a key prerequisite for electric vehicles to completely replace traditional fuel vehicles. The improvement of battery technology, vehicle performance, and cost advantages should be accelerated to increase the market acceptance of electric vehicles. Furthermore, the electricity generation structure determines the energy saving and emission reduction potential of electric vehicles. At present, the high proportion of thermal power in China's power generation mix largely limits the energy and environmental benefits of electric vehicles. Relevant national energy development plans such as increasing the proportion of renewable energy power generation and reducing energy consumption for power generation in coal-fired thermal power plants should be effectively implemented. Finally, taking into account the heterogeneity of electricity generation and supply in various regions, different EV promotion plans or timetables for banning the sale of TFV could be developed for different regions. In regions with a high proportion of renewable energy power generation, such as South China, the time point for the ban on the sale of traditional fuel vehicles could be a bit earlier than in other regions. For regions with a high proportion of coal-fired thermal power, such as North and East China, the large-scale promotion of electric vehicles should be appropriately slowed down, and the promotion plan should be adjusted in time as the proportion of renewable energy power generation increases.

However, there also exist some limitations in the study. First of all, this study selects a passenger car as the research object for comparative analysis and does not consider other components of the road transportation system such as transit buses, sanitation vehicles, and other types of heavy-duty vehicles. In addition, due to the lack of more relevant data, this article focuses on the energy and environmental impact of the battery electric vehicle, while some other types of new energy vehicles are not included in the research scope, such as plug-in hybrid electric vehicles, natural gas vehicles, and fuel-cell electric vehicles. Moreover, the impacts of vehicle manufacturing technology, fuel technology, and battery technology on the life cycle energy consumption and greenhouse gas emission intensities of the vehicles also need to be further supplemented and improved in the future. Finally, in terms of environmental impact, the LCA study in this article focuses solely on greenhouse gas emissions.

Therefore, further systematic and comprehensive LCA studies are needed to assess the wide range of environmental impacts of promoting new energy vehicles.

Data Availability All data generated or analyzed during this study are included in this published article.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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